The Geometry of Misalignment

Angular Distance Between Image-Specific KLT Axes and BT.601 Fixed Axes Across the Kodak Lossless True Color Image Suite

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Abstract

Baetzel (2026a) [4] computed the per-image KLT eigendecomposition of all 24 images in the Kodak Lossless True Color Image Suite, establishing the theoretical decorrelation ceiling at residual correlation r = 0.000. Baetzel (2026b) [5] measured BT.601 decorrelation efficiency against that ceiling and found it ranges from 31.5% to 82.3% across the suite but did not explain the source of that variation. This paper provides the explanation. Both the KLT and BT.601 operate as rotations of the RGB coordinate system, and the difference between them is measurable as an angular distance in three-dimensional color space. We compute the axis-by-axis angular misalignment between each image's optimal KLT rotation and BT.601's fixed rotation for all 24 images, producing three angles per image: the misalignment between PC1 and the Y (luminance) axis, between PC2 and the Cb axis, and between PC3 and the Cr axis. An energy-weighted reduction of these three angles to a single scalar—the standard approach to summarizing multivariate misalignment—produces no predictive power ($R^2 = 0.007$). A multiple regression using all three angles independently explains 57.6% of the variance in BT.601 decorrelation efficiency ($R^2 = 0.576$, $n = 24$). This failure of energy-weighting reveals a counterintuitive geometric reality: misalignment on the minor chrominance axes, which carry less than 5% of total variance, causes more residual redundancy than misalignment on the dominant luminance axis carrying more than 95%. The implication is that transform design strategies prioritizing luminance fidelity while treating chrominance alignment as secondary may be optimizing the wrong axes. Every measurement in this paper is derived from data already published in [4] and [5]; the contribution is not new data but a new geometric relationship between existing variables that has not been quantified in three decades of active use of this benchmark.

1 Introduction

The BT.601 YCbCr transform has served as the decorrelation step in JPEG and its successors since 1992 [1], converting RGB pixel data into a luminance-chrominance representation using fixed coefficients derived from broadcast television luminance characteristics [2] rather than from any property of the image being compressed. The Karhunen-Loève Transform computes an image-specific rotation from each image's covariance structure and achieves residual inter-channel correlation of exactly zero by aligning the transform axes with the data's principal components [3]. The gap between these two—measured per image across the Kodak Lossless True Color Image

Suite in [5]—ranges from 31.5% to 82.3% in decorrelation efficiency, a nearly threefold variation that suite-averaged reporting has historically concealed.

That variation was documented but not explained. The per-image gap analysis [5] identified the Cb–Cr chroma pair as the dominant residual (mean $|r| = 0.616$) but did not address why BT.601 performs well on certain images and poorly on others. The dimensionality tier analysis offered partial structure—near-degenerate images averaged 39.3% gap while three-dimensional images averaged 51.6%—but the relationship between dimensionality and efficiency was not monotonic, and the tiers themselves are aggregate shape descriptors that do not capture the orientation of the color distribution in RGB space.

This paper addresses the orientation question directly. Both BT.601 and the KLT are rotations of the three-dimensional RGB coordinate system. BT.601 applies the same rotation to every image; the KLT computes the optimal rotation per image from the covariance matrix. The angular distance between these two rotations—decomposed axis by axis into the misalignment between PC1 and Y, PC2 and Cb, PC3 and Cr—is a geometric quantity that has never been computed or published for the Kodak suite, despite being derivable from data that has been publicly available since the early 1990s.

Since its standardization in 1982 and adoption into JPEG in 1992, BT.601 has functioned as infrastructure rather than as a testable hypothesis about color-space geometry. Researchers have optimized quantization tables, entropy coding schemes, and spatial transforms around the YCbCr representation without examining whether the representation itself is geometrically well-suited to the data passing through it on a per-image basis. The per-image baselines established in the first two papers of this series make that examination possible, and the present paper carries it out. A companion paper [6] establishes that θ_2 and θ_3 are not independent: across all 24 images, the two chrominance misalignment angles co-vary at $r = 0.9993$, constraining the misalignment space to two geometric degrees of freedom rather than three.

2 Theoretical Framework

A color image with three channels can be represented as a cloud of points in three-dimensional RGB space, where each pixel is a coordinate [R, G, B]. The covariance matrix of this point cloud defines an ellipsoid whose principal axes correspond to the eigenvectors of the matrix and whose extent along each axis is proportional to the square root of the corresponding eigenvalue. The KLT rotates the coordinate system to align with these principal axes, producing transformed channels with zero mutual correlation [3].

BT.601 performs a different rotation, defined by the coefficient matrix [2]:

$$
\begin{aligned}
Y &= 0.299R + 0.587G + 0.114B \\
Cb &= -0.169R - 0.331G + 0.500B + 128 \\
Cr &= 0.500R - 0.419G - 0.081B + 128
\end{aligned}
$$

Each row of this matrix defines a direction in RGB space. The Y row defines the luminance axis; the Cb and Cr rows define the two chrominance axes. These three directions are fixed across all images. When the covariance ellipsoid of a particular image happens to align with these fixed directions, BT.601 approximates the KLT closely and decorrelation is efficient. When the ellipsoid is rotated away from these directions, the fixed transform leaves residual correlation in proportion to the angular distance between the two axis systems.

The angular misalignment between a KLT eigenvector and its corresponding BT.601 axis is computed as the arccosine of the absolute value of their dot product, both vectors normalized to unit length. This produces an angle in degrees ranging from 0° (perfect alignment) to 90° (orthogonal axes). Three such angles fully characterize the axis-by-axis relationship between the two rotations for a given image: PC1↔Y, PC2↔Cb, and PC3↔Cr.

3 Methodology

All measurements in this paper are derived from two previously published datasets. The per-image eigendecomposition (eigenvalues, eigenvectors, variance explained) comes from the PCA characterization [4]. The per-image BT.601 residual correlations (Y–Cb, Y–Cr, Cb–Cr) and decorrelation efficiency come from the gap analysis [5]. No additional image processing or feature extraction was performed; the analysis operates entirely on the statistical outputs of the prior two papers.

3.1 Angular Misalignment Computation

For each of the 24 images, the three eigenvectors from [4] and the three row vectors of the BT.601 coefficient matrix were normalized to unit length. The angular distance between each corresponding pair was computed as:

$$\theta_i = \arccos(|e_i \cdot b_i|)$$

where e_i is the i-th KLT eigenvector and b_i is the i-th BT.601 row vector. The absolute value of the dot product accounts for sign ambiguity in eigenvector direction, which is arbitrary and carries no geometric meaning. This produces three angles per image: θ_1 (PC1↔Y), θ_2 (PC2↔Cb), and θ_3 (PC3↔Cr). The axis correspondence follows eigenvalue ordering: PC1, which captures the largest share of variance, is paired with Y (luminance); PC2 and PC3, in descending order, are paired with Cb and Cr respectively. This pairing is empirically validated by the angular distances themselves—PC1↔Y misalignment is consistently low across the suite (21.89°–32.93°), confirming that the dominant variance axis in natural images aligns with the luminance direction.

3.2 Predictive Analysis

Two models were tested against the decorrelation efficiency measurements from [5]. The first is an ordinary least-squares multiple regression using all three angles as independent variables and decorrelation efficiency as the dependent variable (n = 24). The second is a single-variable model

using an energy-weighted composite angle, computed as the variance-weighted sum of the three angular misalignments:

$$\theta_weighted = \Sigma\ (\lambda_i\ /\ \Sigma\lambda)\ \cdot\ \theta_i$$

where λ_i are the eigenvalues from [4]. The energy-weighted model assigns importance to each axis in proportion to the variance it carries, reflecting the intuition that misalignment on a high-energy axis should matter more than misalignment on a low-energy axis.

4 Results

4.1 Per-Image Angular Misalignment

Table 1 presents the angular misalignment between each image's KLT axes and the corresponding BT.601 axes for all 24 images in the suite, alongside the decorrelation efficiency reported in [5].

Image	θ_1 (PC1↔Y)	θ_2 (PC2↔Cb)	θ_3 (PC3↔Cr)	Total θ	Eff. %	Tier
kodim01	27.27°	67.54°	65.90°	160.71°	53.0%	S 1-D
kodim02	29.20°	66.19°	64.00°	159.39°	37.3%	3-D
kodim03	22.54°	32.21°	29.76°	84.51°	43.2%	3-D
kodim04	32.37°	79.75°	76.95°	189.07°	33.1%	W 1-D
kodim05	26.21°	47.05°	45.11°	118.37°	69.2%	W 1-D
kodim06	26.67°	61.29°	58.88°	146.84°	31.5%	N-D
kodim07	30.80°	56.08°	54.17°	141.05°	75.6%	W 1-D
kodim08	26.98°	47.11°	45.12°	119.21°	59.2%	S 1-D
kodim09	24.75°	32.68°	29.93°	87.36°	48.5%	W 1-D
kodim10	25.72°	50.02°	47.90°	123.64°	56.6%	S 1-D
kodim11	28.54°	66.28°	64.45°	159.27°	61.6%	W 1-D
kodim12	27.69°	56.80°	55.14°	139.63°	61.9%	S 1-D
kodim13	30.40°	23.89°	21.98°	76.27°	78.3%	N-D
kodim14	21.89°	39.41°	35.38°	96.68°	36.5%	2-D
kodim15	27.37°	71.61°	69.91°	168.89°	57.7%	S 1-D
kodim16	27.29°	31.85°	29.76°	88.90°	54.6%	S 1-D
kodim17	28.67°	36.30°	34.45°	99.42°	55.2%	N-D
kodim18	27.34°	40.32°	36.81°	104.47°	32.1%	W 1-D
kodim19	28.98°	33.54°	31.48°	94.00°	46.3%	S 1-D
kodim20	30.87°	24.71°	22.77°	78.35°	63.5%	N-D
kodim21	30.19°	40.00°	38.13°	108.32°	53.2%	W 1-D
kodim22	32.93°	39.90°	37.97°	110.80°	82.3%	W 1-D
kodim23	29.61°	46.27°	44.30°	120.18°	64.9%	3-D

kodim24	29.30°	32.07°	30.16°	91.53°	75.2%	N-D

Table 1. Per-image angular misalignment between KLT eigenvectors and BT.601 axes. Tier abbreviations: S 1-D = Strongly one-dimensional, W 1-D = Weakly one-dimensional, N-D = Near-degenerate, 3-D = Three-dimensional, 2-D = Two-dimensional.

The PC1↔Y misalignment is remarkably stable across the suite, ranging from 21.89° (kodim14) to 32.93° (kodim22) with a standard deviation of 2.70°. This narrow spread reflects the fact that PC1 is always approximately luminance-aligned in natural photographic images—the dominant variance axis typically runs close to the luminance diagonal because RGB channels are positively correlated along a direction that BT.601's Y coefficients approximate.

The chrominance axes tell a different story. PC2↔Cb ranges from 23.89° (kodim13) to 79.75° (kodim04), and PC3↔Cr ranges from 21.98° (kodim13) to 76.95° (kodim04)—each spanning more than 54° across the suite. The chrominance axes of BT.601 are fixed directions in RGB space, but the actual minor axes of the covariance ellipsoid rotate substantially from image to image depending on the spectral content of the scene, the dominant illuminant, and the film stock's dye sensitivities. BT.601's assumption that there exists a single pair of chrominance directions adequate for all images is contradicted by the data: the optimal chrominance axes vary by more than 54° across a 24-image suite.

4.2 Three-Axis Regression

Table 2 presents the bivariate correlations between each angular misalignment and decorrelation efficiency, followed by the multiple regression result.

Predictor	r	R^2	p
θ_1 (PC1↔Y) vs. Efficiency	+0.430	0.185	< 0.05
θ_2 (PC2↔Cb) vs. Efficiency	−0.283	0.080	n.s.
θ_3 (PC3↔Cr) vs. Efficiency	−0.258	0.066	n.s.
Multiple regression (θ_1, θ_2, θ_3)	R = 0.760	0.576	< 0.001
Energy-weighted θ	+0.086	0.007	n.s.

Table 2. Angular misalignment as predictor of BT.601 decorrelation efficiency (n = 24). The multiple regression uses all three angles as independent variables. The energy-weighted model collapses the three angles to a single variance-weighted scalar.

The three-axis multiple regression explains 57.6% of the variance in BT.601 decorrelation efficiency across the suite. The individual bivariate correlations reveal an asymmetry: θ_1 alone reaches $R^2 = 0.185$, while θ_2 and θ_3 individually fall below statistical significance at $R^2 = 0.080$ and 0.066 respectively. The combined model captures substantially more than any axis alone, indicating that the three misalignment dimensions jointly contribute to decorrelation performance—the chrominance axes add predictive power only when considered alongside the luminance axis.

4.3 The Energy-Weighting Failure

The energy-weighted composite angle produces $R^2 = 0.007$—effectively zero predictive power. This result is initially counterintuitive: weighting each axis by the fraction of total variance it carries should, by standard reasoning, concentrate the metric on the axes that matter most. In practice, it does the opposite. Because PC1 typically captures 85–98% of variance in natural images, the energy-weighted angle collapses to approximately the PC1↔Y misalignment alone, which explains only 18.5% of efficiency variance. The information carried by θ_2 and θ_3—which contribute jointly to the multiple regression's explanatory power—is suppressed to near-invisibility by the weighting.

This is the central quantitative finding of the paper. The chrominance axes carry a small fraction of total image energy—often less than 5% combined—yet misalignment on these axes is more strongly associated with decorrelation efficiency variance than misalignment on the dominant luminance axis. A 5° rotation error on an axis carrying 95% of variance produces less residual correlation than a 50° rotation error on an axis carrying 2% of variance, because the high-energy axis is broad enough that nearby directions still capture most of the spread, while the low-energy axis is narrow enough that even moderate angular error projects variance onto the wrong transformed channel.

4.4 Axis-Specific Residual Correlations

Table 3 reports the bivariate correlations between each angular misalignment and each of the three YCbCr residual correlation pairs.

Angular Misalignment	|Y–Cb| residual	|Y–Cr| residual	|Cb–Cr| residual
θ_1 (PC1↔Y)	−0.540	+0.051	+0.013
θ_2 (PC2↔Cb)	+0.038	+0.504	−0.048
θ_3 (PC3↔Cr)	+0.013	+0.502	−0.050

Table 3. Bivariate correlations between per-axis angular misalignment and per-pair YCbCr residual correlations (n = 24).

The individual angular misalignments present a counterintuitive relationship with the resulting YCbCr residuals. Closer PC1↔Y alignment actually correlates with a higher Y–Cb residual ($r = -0.540$). This occurs because when PC1 aligns well with luminance, the remaining variance is forced onto the minor axes, whose misalignment with Cb and Cr then dictates how the error distributes.

The PC2↔Cb and PC3↔Cr misalignments both correlate with the Y–Cr residual at $r \approx +0.50$, a direct and interpretable relationship: when the minor principal axes diverge from BT.601's chrominance directions, the luminance-chrominance residual increases.

The most critical finding is that the Cb–Cr residual—the dominant source of post-transform redundancy—shows no significant correlation with any single axis misalignment. This absence indicates that the Cb–Cr redundancy is not an orientation problem but is consistent with a structural property of how BT.601 bisects the chroma plane. The structural coupling between the two chrominance axes persists regardless of how the rotation as a whole sits in the RGB cube. While subsequent standards (BT.709, BT.2020) updated the luminance coefficients to reflect evolving display technology, this architectural coupling remains unchanged.

5 Discussion

The measurements in this paper quantify a relationship that compression researchers have had the data to compute since the early 1990s. The Kodak suite has been publicly available for three decades. PCA eigendecomposition of RGB images is a standard operation. The BT.601 coefficient matrix is published in an international standard. Computing the angular distance between the resulting axis systems requires a dot product and an arccosine. The 57.6% explained variance is not a subtle signal requiring sophisticated detection; it is a first-order geometric property of the data that, to our knowledge, has not previously been measured.

The explanation for this gap in the literature is most likely methodological rather than conceptual. BT.601 entered compression practice as a standard, and standards are typically treated as fixed components of a pipeline rather than as variables to be analyzed. Published benchmarks against the Kodak suite report the outputs of compression pipelines—rate-distortion curves, PSNR, SSIM—not the internal performance of the colorspace conversion step in isolation. The per-image PCA baselines in the first paper of this series were themselves the first published eigendecomposition of the suite, indicating that the prerequisite data for the present analysis did not exist in the literature until recently.

The energy-weighting failure carries a specific implication for transform design. If misalignment on the minor axes has a disproportionate impact on decorrelation performance, then a transform optimized to match the luminance axis as closely as possible while treating the chrominance axes as secondary would underperform a transform that distributes its alignment effort more evenly across all three axes. BT.601 was designed to extract luminance cleanly, with chrominance defined as the residual. The data indicates that this priority ordering produces a characteristic failure mode: the luminance axis is well-matched (narrow θ_1 range, 21.89°–32.93°), but the chrominance axes are left uncontrolled (θ_2 range 23.89°–79.75°), and it is exactly this uncontrolled variation that drives the decorrelation efficiency differences across the suite.

The 42.2% of efficiency variance not explained by angular misalignment likely stems from the condition number and ellipsoid shape metrics established in Baetzel (2026a) [4]. A model incorporating these alongside the orientation angles would likely capture additional variance, though it would introduce collinearity with the dimensionality metrics already reported. This interaction is not modeled here; the present analysis isolates the contribution of orientation alone.

The orthogonal constraint documented in [6] further clarifies the model specification. Because θ_2 and θ_3 co-vary at r = 0.9993 with slope ≈ 1.004, they are functionally a single degree of freedom. The appropriate specification is therefore a two-predictor model using θ_1 (luminance misalignment) and a single chrominance angle φ, reducing the geometric degrees of freedom from three to two without loss of explanatory power. The three-axis regression reported in Section 4.2 remains diagnostic, but the true parameter space is two-dimensional.

6 Conclusion

The angular distance between each image's KLT-optimal axes and BT.601's fixed axes explains 57.6% of the per-image variance in decorrelation efficiency across the Kodak Lossless True Color Image Suite. The chrominance axes—which carry the least energy—exhibit the widest misalignment variation (>54° range) and contribute disproportionately to the prediction, a finding confirmed by the failure of energy-weighted angular composites to retain predictive power. These measurements are derived entirely from data published in [4] and [5], require no image-level processing to reproduce, and are provided as executable scripts operating on the machine-readable outputs of the prior two papers.

References

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[2] ITU-R, "Studio encoding parameters of digital television for standard 4:3 and wide-screen 16:9 aspect ratios," Recommendation BT.601-7, 2011.

[3] Watanabe, S. "Karhunen–Loève Expansion and Factor Analysis: Theoretical Remarks and Applications." pp. 635–660, 1965.

[4] J. Baetzel, "Statistical Characterization of Inter-Channel Redundancy Structure in the Kodak Lossless True Color Image Suite: Per-Image Principal Component Decomposition of PCD0992," 2026. https://github.com/PearsonZero/kodak-pcd0992-statistical-characterization

[5] J. Baetzel, "Per-Image Decorrelation Efficiency of the BT.601 Fixed Transform: Measured Against KLT Ceiling Across the Kodak Lossless True Color Image Suite," 2026. https://github.com/PearsonZero/kodak-pcd0992-bt601-decorrelation-gap

[6] J. Baetzel, "The Orthogonal Constraint on Chrominance Axis Misalignment in the Kodak Lossless True Color Image Suite," 2026. https://github.com/PearsonZero/kodak-pcd0992-orthogonal-constraint

Source

Benchmark images: Kodak Lossless True Color Image Suite (r0k.us/graphics/kodak/). 24 images, 768×512 / 512×768, 24-bit RGB, lossless PNG. Baseline characterization: [4]. Gap measurements: [5]. Analysis: Python 3, NumPy. All metrics computed from published per-image JSON outputs of the prior two papers.